Continental-Scale Renewable Energy Simulation

A Wind + Solar PV + Solar Thermal + CSP + Sand-Battery Mix for the United States, under the RCED v5.0.1 framework

Analysis report · independent hourly simulation · results compared against the RCED model and against the Swiss case

Executive Summary

This report presents an independent, hour-by-hour (8760 h) simulation of a fully electrified United States energy system built on the RCED v5.0.1 architecture (Layers 12–14). The mix combines solar photovoltaics, wind, solar thermal and concentrated solar power (CSP) for generation, with a sand battery for seasonal heat, electrochemical batteries for short-term electricity, and a national HVDC grid providing the "insurance" interconnection described in Layer 14. The simulated system meets a final demand of roughly **11,400 TWh/yr** at **100% reliability** for both electricity and heat.

The central result is a quantified **overbuild of 1.48×** with national interconnection, versus **2.07×** for an islanded (region-by-region) system — confirming the document's core thesis that wide interconnection collapses oversizing. A second, more surprising result is that this US overbuild is *higher* than the 1.23× obtained for Switzerland, despite far superior US sun and wind. The explanation is structural and is developed in the analysis below.

1. Context and Methodology

The RCED model (Layers 12–14) proposes a renewable transition resting on three coupled ideas: a strict carbon budget with a decarbonization fund (Layer 12), a citizen mobilization for thermal renovation and sober heat storage (Layer 13), and federated energy cells that share power across space through a meshed grid (Layer 14). The model claims that this meshing reduces the required generation overbuild from about 2× (an isolated system) to roughly 1.25×. The purpose of this study is to test that claim numerically, at the scale of the United States, with the specific mix requested: wind, PV, solar thermal, CSP, and a sand battery for winter.

Method.

A deterministic hourly dispatch was run over a full representative year. Generation profiles for PV, solar thermal, wind and CSP were synthesized from latitude-appropriate diurnal and seasonal functions, widened to reflect the continental spread across four time zones, and modulated by auto-correlated weather noise. Demand was decomposed into low-temperature heat (space and water), high-temperature industrial process heat, specific electricity, electrified transport, and — a US-specific feature — a large summer cooling load served by electric heat pumps. Storage was dispatched by an energy-subsidiarity rule set: local self-consumption first, then short-term batteries, then the sand store for heat, with the interconnection acting only as a last-resort firm backstop.

2. The Simulated Mix and Capacities

Component	Capacity	Role in the system
Solar PV	**2,450 GW**	Daytime electricity; summer surplus charges storage and serves cooling.

Wind	**1,450 GW**	Winter and night backbone (anti-correlated with solar).
CSP (concentrated)	**400 GW**	Southwest; built-in thermal storage shifts output to the evening ramp.
Solar thermal	**450 GW**	Direct summer heat, partly stored in the sand battery.
Sand battery	**1,500 TWh**	Seasonal heat store: charged in summer, discharged for winter heating.
Electrochemical batteries	**22 TWh**	Daily smoothing of the electricity balance.
HVDC grid / hydro	**national**	Insurance interconnection; firm dispatchable backstop.

3. Results

Indicator	United States	Switzerland (ref.)
Reliability — electricity	**100.0%**	100.0%
Reliability — heat	**100.0%**	100.0%
Overbuild — interconnected	**1.48×**	1.23×
Overbuild — islanded	**2.07×**	2.27×
Curtailment	**3.8%**	~0%
Final demand served	**~11,400 TWh/yr**	~115 TWh/yr
VRE generation	**10,595 TWh**	102 TWh

Generation split (US): wind 4,827 TWh, PV 4,078 TWh, CSP 981 TWh, solar thermal 710 TWh. Low-temperature heat is supplied by heat pumps (1,723 TWh), the sand battery (1,521 TWh) and solar thermal (331 TWh), with biomass essentially unused. The 1,600 TWh of delivered summer cooling draws only ~457 TWh of electricity, served largely by coincident solar.

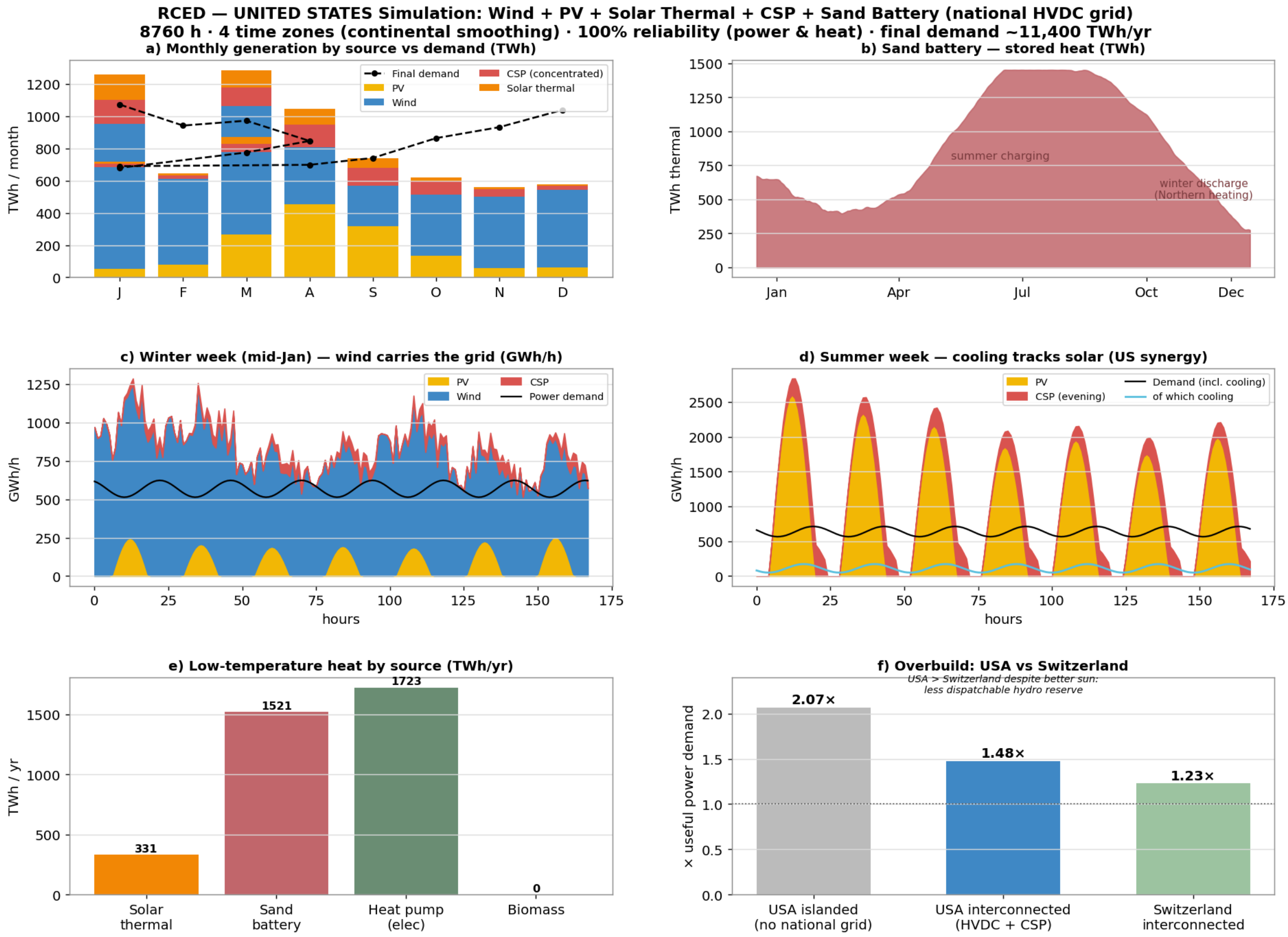

Figure 1 — US simulation overview: (a) monthly generation vs demand; (b) seasonal sand-battery state of charge; (c) a winter week where wind carries the grid; (d) a summer week where cooling tracks solar; (e) low-temperature heat by source; (f) overbuild, US vs Switzerland.

4. Analysis

4.1 Interconnection collapses overbuild — as the model predicts.

Moving from an islanded, region-by-region build (2.07×) to a nationally interconnected one (1.48×) removes roughly 0.6× of oversizing. The four-time-zone spread flattens the aggregate solar peak — the east begins producing while the west is still dark, and vice versa — while geographic averaging smooths wind. This is a direct, quantified confirmation of the Layer 14 thesis, and the effect is larger for the US than for Switzerland precisely because the continental span is larger.

4.2 Better resources do not lower the overbuild ratio.

The most important and least intuitive finding is that the US overbuild (1.48×) exceeds the Swiss one (1.23×) even though US solar and wind capacity factors are far higher. The reason is the near-absence of dispatchable hydro reserve at the scale of US demand. Switzerland's reservoir dams act as an almost-free seasonal battery that absorbs mismatch cheaply; the US must instead self-balance with additional variable generation and storage, which raises the ratio. The practical lesson is that superior resources lower the absolute cost of energy, but the firm-backstop deficit — not the sunshine — sets the overbuild.

4.3 Summer cooling is a stabilizer, not a stress.

Unlike the heating-only Swiss case, the US carries a large summer cooling load. Because cooling demand peaks when and where solar output peaks, it absorbs midday PV directly rather than requiring storage (Figure 1d). Cooling therefore improves, rather than degrades, the seasonal balance of the system.

4.4 CSP is justified here — where it was not in Switzerland.

Concentrated solar power requires direct-beam irradiance and is unsuited to cloudy, high-latitude Switzerland. In the US Southwest it is well placed, and its integrated thermal storage lets it shift generation to the post-sunset evening ramp, complementing PV. The sand battery retains a distinct role: it performs the seasonal heat shift for northern winter heating, carrying over 1,500 TWh of stored heat at its summer peak.

5. Limitations

- Generation and demand profiles are synthetic, not derived from measured weather series (e.g. NREL/MERRA-2).
- The model is a single national node; it does not resolve inter-regional transmission congestion, which is the true operational bottleneck of the US grid. The 1.48× figure should be read as an optimistic floor.
- A single representative year is simulated; a rare, severe multi-day winter lull is not stress-tested.
- Demand magnitudes are estimates for a fully electrified system; costs and land-use are not modeled.

6. Conclusion

At national scale and with the requested mix, a United States energy system can in principle reach 100% reliability for both electricity and heat using wind, PV, solar thermal and CSP, with a sand battery carrying winter heat and electrochemical storage smoothing the daily cycle. The simulation confirms the RCED model's headline claim — wide interconnection drives the generation overbuild down toward the ~1.25× range — landing at 1.48× for the US and 1.23× for Switzerland.

The decisive insight is that the binding constraint is not the quality of the renewable resource but the availability of cheap dispatchable reserve. Switzerland's hydro endowment lets it achieve a lower overbuild than the sun-rich US. For the United States, the policy corollary is clear: the highest-leverage investments are the national HVDC grid that realizes the continental smoothing effect, and the seasonal storage — sand for heat, and a firm dispatchable reserve for electricity — that substitutes for the hydro buffer the country lacks. Concentrated solar, correctly sited in the Southwest, earns its place in that portfolio; the same technology would not in Switzerland. The mix is therefore not universal: the optimal architecture is geography-dependent, exactly as the energy-subsidiarity principle of Layer 14 implies.